

Kubernetes Resource Limits

This is an important KCNA topic. Let's understand it from **zero level**.

1. What are Resource Limits?

Resource limits tell Kubernetes the maximum amount of CPU and memory that a container is allowed to use.

Think of a computer with limited resources:

Computer

|

|— CPU

└— Memory (RAM)

Your Pods use these resources.

Kubernetes allows you to set a **limit** so one container doesn't use everything.

2. Why Do We Need Resource Limits?

Imagine you have:

Node

|

|— Pod A

|— Pod B

└— Pod C

Suppose Pod A suddenly uses a huge amount of memory.

Without limits:

Pod A → uses almost all RAM ❌

Pod B → affected

Pod C → affected

This can cause problems for other applications.

With a limit:

Pod A

|

└─ Memory Limit = 512Mi

Pod A cannot use more than the configured memory limit.

Easy memory trick:

 **Resource Limit = Maximum allowed resource**

3. Two Main Resources

For KCNA, remember these two:

CPU

CPU controls **processing power**.

Example:

cpu: 500m

Memory

Memory means **RAM**.

Example:

memory: 512Mi

4. Resource Requests vs Resource Limits

This is VERY important because they are often confused.

Request

The amount of resource a container asks Kubernetes to reserve for it.

Limit

The maximum amount of resource the container is allowed to use.

Think about a hotel room:

Request = "I need this much space."

Limit = "I cannot use more than this much space."

5. Example

resources:

requests:

cpu: "250m"

memory: "256Mi"

limits:

cpu: "500m"

memory: "512Mi"

Here:

Request

CPU → 250m

Memory → 256Mi

Limit

CPU → 500m

Memory → 512Mi

So Kubernetes tries to provide at least the requested resources, while the container's usage is constrained by the limits.

6. What Does m Mean in CPU?

You will see:

500m

The m means **millicpu**.

1000m = 1 CPU

500m = 0.5 CPU

250m = 0.25 CPU

100m = 0.1 CPU

Easy memory:

1000m = 1 CPU

7. Memory Units

You may see:

Mi

Gi

For example:

memory: "512Mi"

means approximately **512 MiB** of memory.

Common examples:

256Mi

512Mi

1Gi

2Gi

8. Complete Pod Example

apiVersion: v1

kind: Pod

metadata:

name: nginx

spec:

containers:

- name: nginx

image: nginx

resources:

requests:

cpu: "250m"

memory: "256Mi"

limits:

cpu: "500m"

memory: "512Mi"

The important part is:

resources:

requests:

cpu: "250m"

memory: "256Mi"

limits:

cpu: "500m"

memory: "512Mi"

9. What Happens When CPU Limit is Reached?

Suppose:

CPU Limit = 500m

The container tries to use more CPU.

Kubernetes **throttles** the CPU usage.

In simple words:

The container's CPU usage is restricted.

10. What Happens When Memory Limit is Exceeded?

This is different from CPU.

Suppose:

Memory Limit = 512Mi

The container tries to use much more memory.

If it exceeds the memory limit, the container can be **terminated by the kernel due to out-of-memory (OOM)**.

You may see:

OOMKilled

Remember:

CPU → Throttled

Memory → Can be OOMKilled

★ This is useful for KCNA.

11. Why Are Requests Important?

Suppose you have:

Node:

4 CPU

Pod A requests:

2 CPU

Pod B requests:

1 CPU

Pod C requests:

2 CPU

Kubernetes looks at the **resource requests** when deciding whether a Pod can fit on a Node.

Pod C may not fit if insufficient requested CPU remains.

This is one of the jobs of the **Scheduler**.

12. Resource Requests and Scheduler

Remember your Kubernetes architecture:

Pod

|



Scheduler

|



Checks Node resources

|



Places Pod on suitable Node

The Scheduler considers resource requests when making scheduling decisions.

13. Easy Real-Life Example 🍕

Imagine a pizza shop has:

10 slices

Three customers request:

Customer A → 3 slices

Customer B → 2 slices

Customer C → 4 slices

Total requested:

$$3 + 2 + 4 = 9$$

The shop can handle it.

But if another customer requests:

5 slices

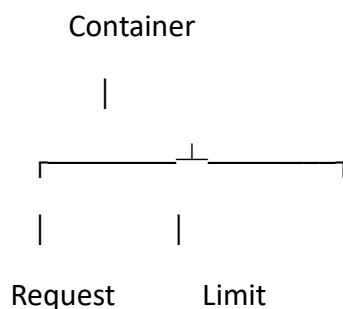
there may not be enough available capacity.

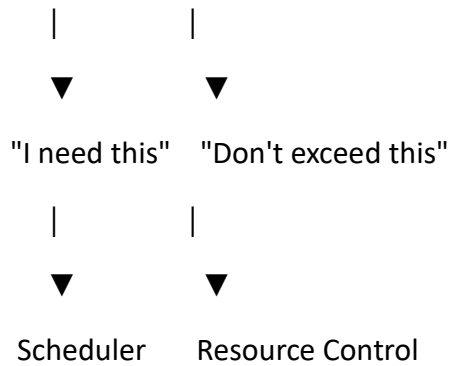
Similarly, Kubernetes considers **resource requests** when deciding whether a Pod can fit on a Node.

14. Resource Limit vs Request

Request	Limit
Minimum/desired resource for scheduling	Maximum resource allowed
Used by Scheduler	Controls container usage
Helps decide where Pod can run	Prevents excessive usage
Example: 250m CPU	Example: 500m CPU

15. Easy Diagram





16. Quick KCNA Revision ★

Resource Request

📌 What the container asks for / reserves for scheduling.

Resource Limit

📌 Maximum amount of CPU or memory the container can use.

CPU

1000m = 1 CPU

500m = 0.5 CPU

Memory

512Mi

1Gi

If CPU limit is reached:

➡ CPU can be **throttled**.

If memory limit is exceeded:

➡ Container can be **OOMKilled**.

Easy Memory Trick 🧠

REQUEST = "I NEED this much." 📦

LIMIT = "I CANNOT USE more than this." 🚧

Most important KCNA sentence:

Requests help Kubernetes decide where a Pod can be scheduled, while limits define the maximum CPU and memory a container can use.